📊 Data Analyst Project Report

Project Title: Sales Dashboard for Regional Performance
Author: K Aditya

1. Executive Summary

This project simulates a corporate-grade, end-to-end data analytics workflow for BMW's global sales data (2018–2025). The goal is to consolidate dispersed datasets into a unified analytics platform, enabling leadership to track performance trends, measure EV adoption, and compare regional market share.

The final deliverable is an interactive **Power BI dashboard** supported by SQL queries, Power Query transformations, and DAX measures, designed to provide actionable insights for strategic decision-making.

2. Business Problem Statement

BMW's global sales data is fragmented across multiple sources, creating challenges in:

- Tracking global and regional performance trends.
- Measuring EV penetration across markets.
- Benchmarking regional market share.
- Supporting timely resource allocation and strategic planning.

Impact if unresolved: slower market response, inefficient resource allocation, missed EV adoption opportunities, and reduced competitiveness in the premium automotive sector.

3. Methodology

Stage	Tools Used	Key Activities
Data Loading	Power Query Imported raw dataset	initial inspection
Exploratory Data Analysis (EDA)	Power BI, SQL	Identified missing values, outliers, distribution patterns
Data Cleaning	Power Query	Removed duplicates, handled nulls, standardized formats
SQL Queries	PostgreSQL	Created tables, ran joins, aggregations, trend analysis
Dashboard Creation	Power BI + DAX	Built KPIs, charts, slicers; calculated YoY, MoM growth
Reporting	Power BI, PPT	Summarized findings, prepared recruiter-friendly presentation

4. Dashboard Features

- **KPIs:** Total Revenue, Units Sold, YoY Growth, MoM Growth.
- **EV Trends:** BEV penetration across regions.
- **Market Share:** Premium segment share by geography.
- **Regional Performance:** Europe, China, USA, Rest of World.
- **Field Parameters:** Dynamic filtering for flexible analysis.

5. Key Insights

- **Revenue Growth:** Consistent YoY increase in Europe and China, stagnation in USA.
- **EV Adoption:** China leads in BEV penetration, Europe shows steady growth, USA lags.
- **Premium Market Share:** Europe dominates, but China is rapidly closing the gap.
- **Regional Trends:** Rest of World shows volatility, requiring closer monitoring.

6. Business Recommendations

- **Accelerate EV adoption in USA** with targeted incentives and marketing.
- **Reallocate resources** toward China to capitalize on rapid growth.
- **Strengthen premium positioning** in Europe to maintain leadership.
- **Develop risk mitigation strategies** for volatile Rest of World markets.

7. Deliverables

- **Dataset:** `bmw_global_sales_2018_2025.csv`
- **SQL Scripts:** Database creation, queries, aggregations.
- **Power BI Dashboard:** Interactive visuals with DAX measures.
- **Report & Presentation:** Structured findings and recommendations.

8. Repository & Usage

git clone https://github.com/Kaditya9/Sales_Dashboard_for_Regional_Performance_data_analysis-SQL-PowerQuery-PowerBI.git cd Sales_Dashboard_for_Regional_Performance_data_analysis-SQL-PowerQuery-PowerBI

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- Open dataset: `bmw_global_sales_2018_2025.csv`
- Load into PostgreSQL SQL Server
- Connect SQL Database to Power BI
- Apply Power Query transformations
- Use DAX measures (`DAX_used_for_Sales_Dashboard_for_Regional_Performance`)
- View dashboard: `Sales_Dashboard_for_Regional_Performance.pdf`

9. License

MIT — free to use, fork, and share.

10. About the Author

👨‍💻 **K Aditya** — Data Analyst passionate about breaking down complex data topics into practical, job-ready projects.